Review Problem 18

- A RISC machine is shown to increase the instructions in a program by a factor of 2. When is this a good tradeoff?

Halve or more CPI * clockrate

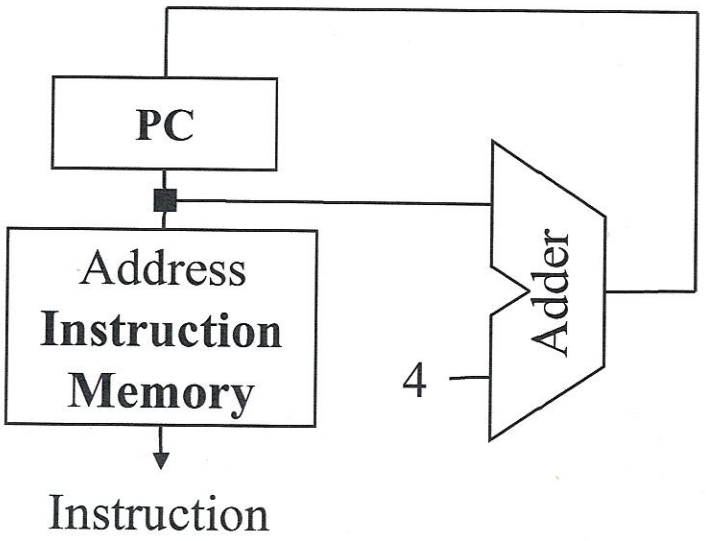
Ex. Code bloat is on infrequently executed code

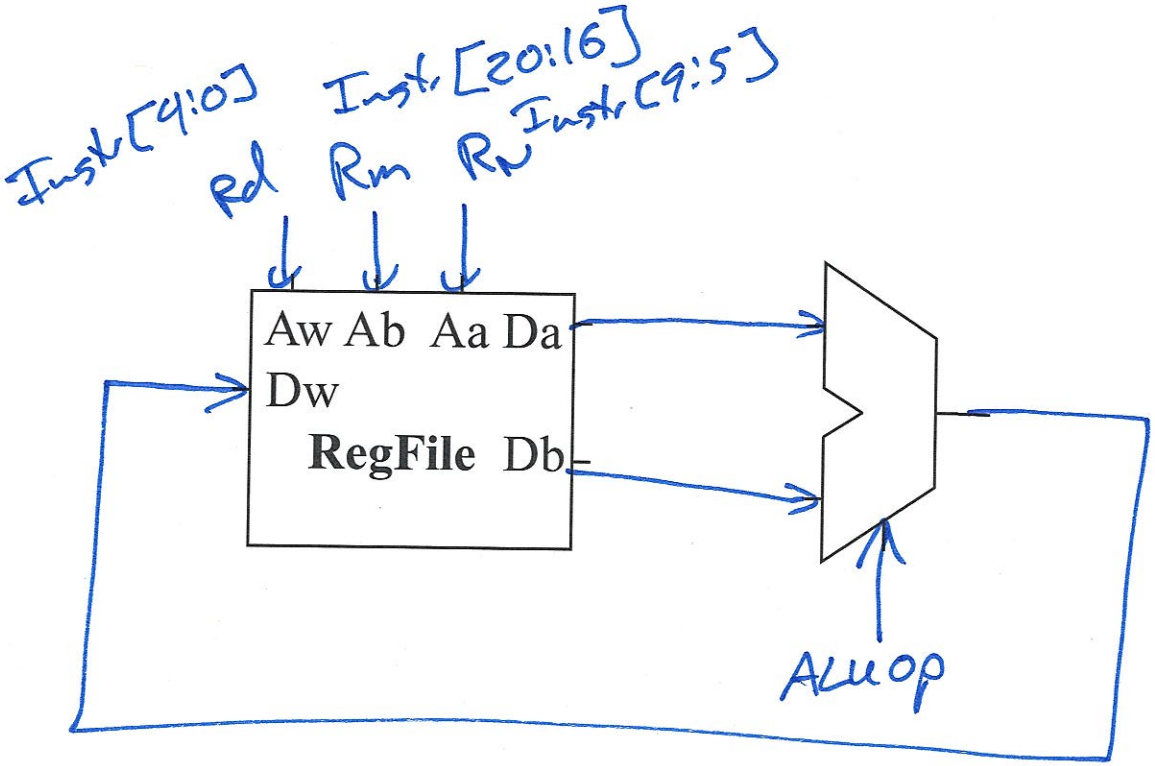
Run multiple instr at the same time

Simplify cpu, smaller.

Add/Subtract Datapath

Reg[Rd] = Reg[Rn] op Reg[Rm]





Load RTL

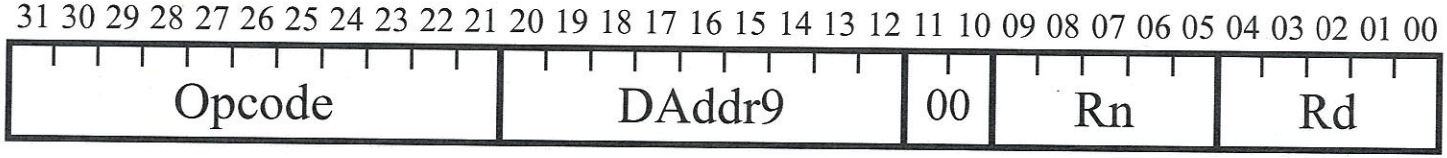
Load Instruction: `LDUR Rd, [Rn, DAddr9]`

Instruction = Mem[PC];

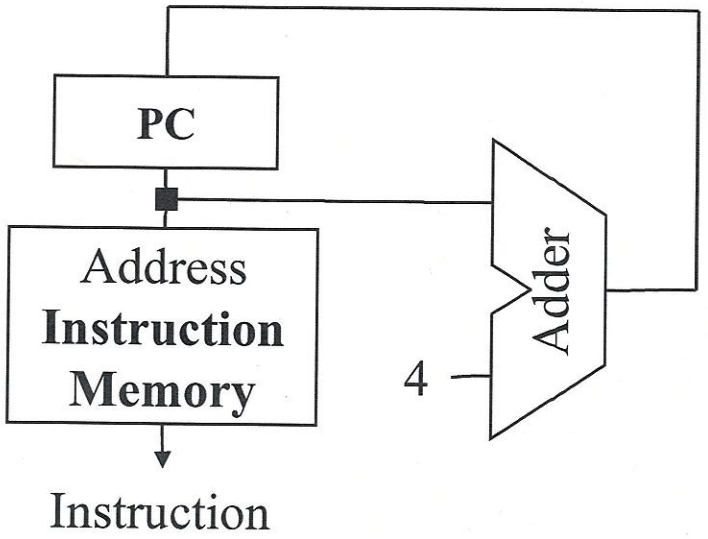
Addr = Reg[Rn] + SignExtend(DAddr9);

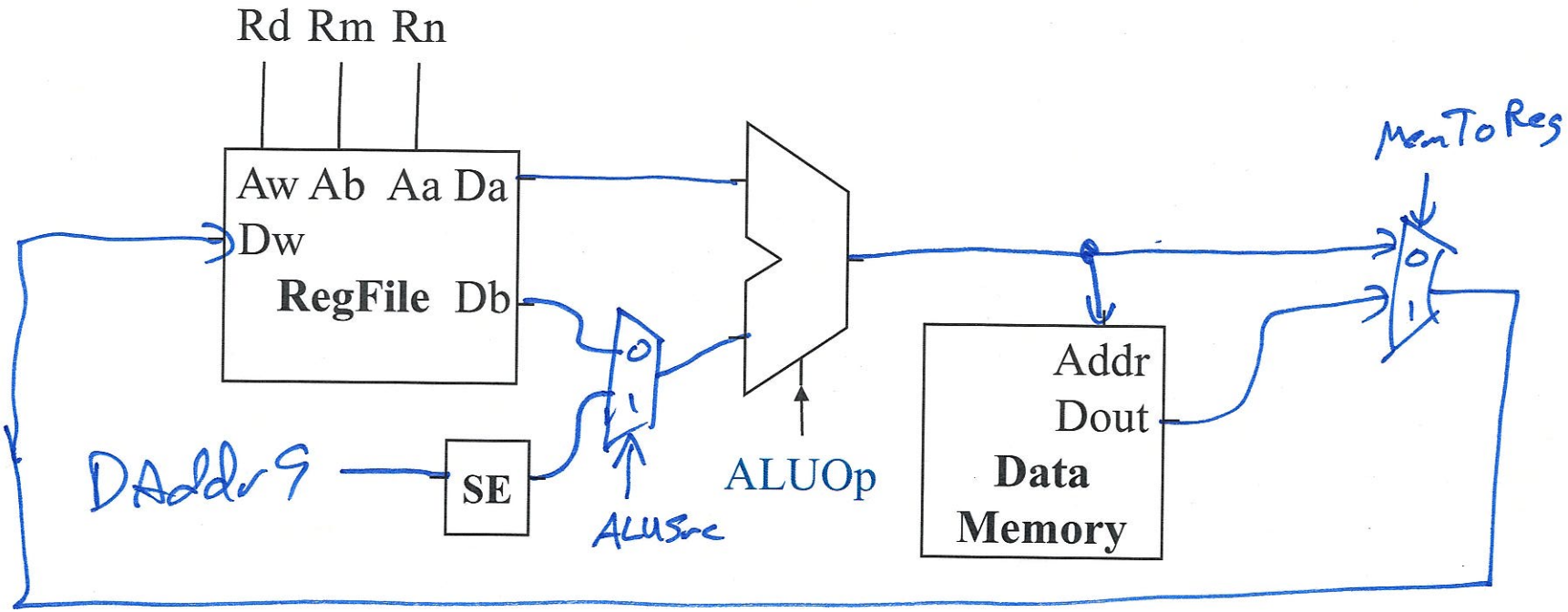
Reg[Rd] = Mem[Addr];

PC = PC+4;



Datapath + Load





Store RTL

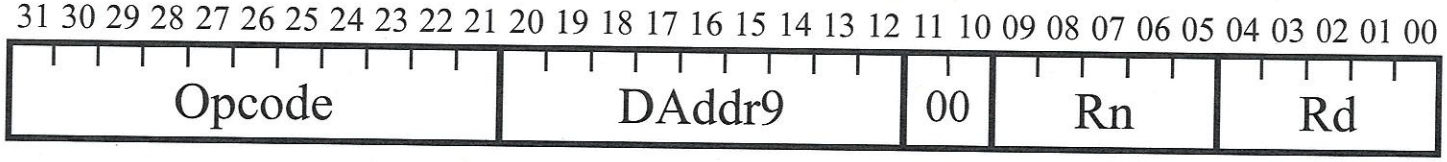
Store Instruction: `STUR Rd, [Rn, DAddr9]`

Instruction = MEM[PC];

Addr = Reg[Rn] + SE(DAddr9);

Mem[Addr] = Reg[Rd];

PC = PC+4;



Datapath + Store

